

ENPH334/PHYS334
Assignment/Lab Project

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Lab Section: Tuesday Group

Group/Station Number 12

Lab Project: High-pass Unity Gain Second Order Sallen-Key Butterworth Filter

TITLE: Analog Lab Project 1 – Designing a High Pass Butterworth Filter.

Declaration:

I/We verify that this assignment or lab project report is my/our own individual work and may include equations and/or diagrams from the ENPH334, PHYS334 notes and instruction sheets. When material has been obtained from an online search, mention is made of this, and the appropriate reference noted.

Signature: NathanielPacey Noah Kriss

[Note: Both partners must sign for lab project report]

Date: October 29, 2022

1 Background Information

The team was assigned to build a High-pass Unity Gain Second Order Sallen-Key Butterworth Filter with a 3dB cut-off Frequency of 5500 Hz. To design this circuit, the team had to understand the exact definitions of such a filter.

1.1 First Order High Pass RC Networks

First the team must understand what makes up a High-pass filter. A high pass RC filter consists of a voltage source in series with a capacitor(s) and a resistor(s). Note that the resistor(s) are in parallel to the output voltage as shown below in Figure 1.

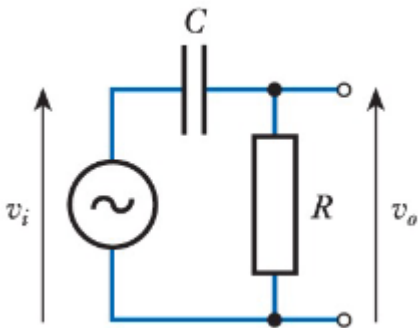


Figure 1. High Pass RC Network Example [1].

Note that for a first order High Pass RC network the Magnitude and Phase of the Voltage Gain can be given by Equations 1 and 2 below.

$$Gain = \left| \frac{v_o}{v_i} \right| = \frac{1}{\sqrt{1 + \left(\frac{1}{\omega CR} \right)^2}} \quad (1)$$

$$Phase = \angle \frac{v_o}{v_i} = - \arctan \left(\frac{1}{\omega CR} \right) \quad (2)$$

And when a fall in Gain of 3dB occurs this represents halving of power. This can be described by Equation 3 and 4.

$$\frac{1}{\omega CR} = 1 \quad (3)$$

$$Gain = \frac{1}{\sqrt{1+1}} = \frac{1}{\sqrt{2}} \quad (4)$$

Note that this is the point of the frequency response know as the cutoff frequency as described by Equation 5 and 6.

$$\omega_c = \frac{1}{CR} \quad (5)$$

$$f_c = \frac{\omega_c}{2\pi} = \frac{1}{2\pi CR} \quad (6)$$

These relations for a first order High-Pass RC Network can be more easily visualized in Bode Diagrams as shown below in Figure 2.

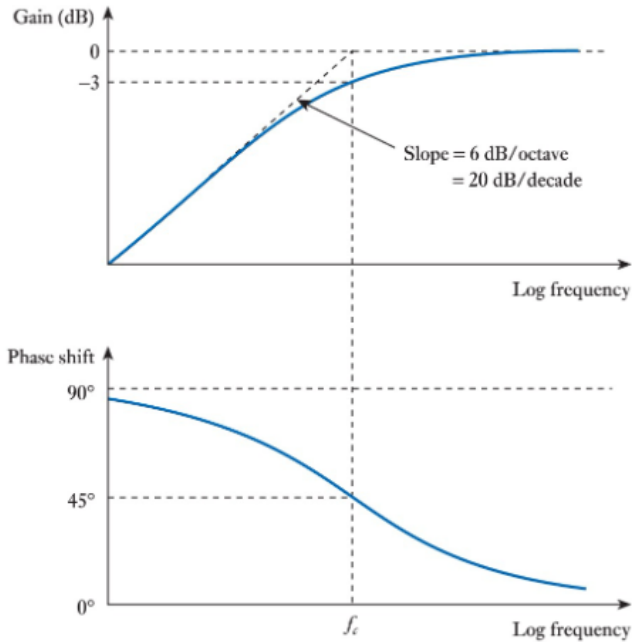


Figure 2. 1st Order High Pass RC Network Bode Plots [1].

1.2 Second Order High Pass RC Networks

A second Order High Pass filter can be constructed using two first order RC filters in parallel as shown below in Figure 3.

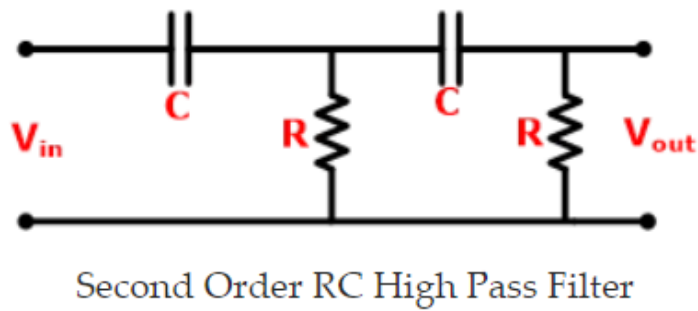


Figure 3. 2nd Order High-Pass RC Network [2].

The comparison from a 1st Order High Pass Filter to a 2nd Order High Pass Filter can be seen below in Figure 3.

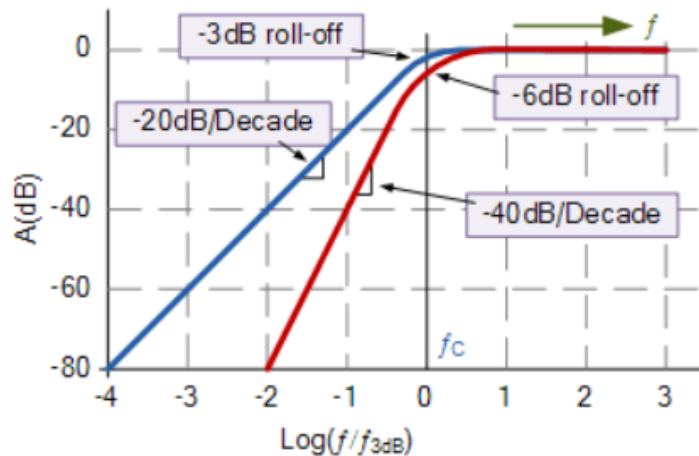


Figure 4. 1st Order vs 2nd Order Bode Plot of Gain vs Frequency Response.

Note that the High Pass filter has a -40 dB/Decade Slope with a -6dB roll-off, while the First Order circuit has a -3dB roll-off with a -20dB/Decade slope. This is because the slope of a second order filter will always be twice the slope of the first order counterpart.

1.3 2nd Order Sallen-Key Butterworth Filter Design

A Butterworth Filter is a type of active filter designed to have a flat frequency response in the pass band region of the frequency plot [2]. A comparison of filters can be seen below in Figure 5, note that these plots are for a Low Pass Network.

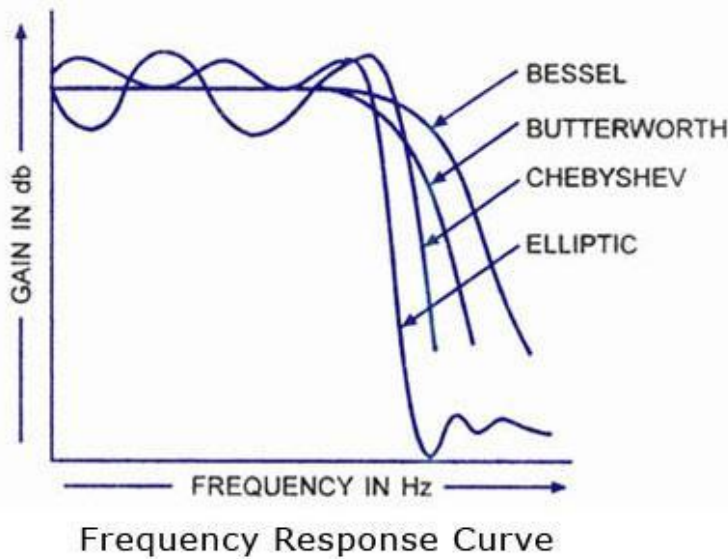


Figure 5. Comparison of Active Filter Types for a Low Pass Network [3].

The attributes of each filter can be seen below in Table 1.

Table 1. Attributes of 2nd Order Filters [4].

Filter Type	Q (2 nd Order)	Roll-off	Response to a step
Butterworth	0.7	-40 dB/decade	Moderate overshoot
Chebyshev	>1	Steeper than Butterworth Ripple/peak in pass band	Large overshoot
Bessel	0.5	Shallower than Butterworth	No overshoot (critical damping)

As well the general circuit design for a Sallen-Key Butterworth High Pass filter can be seen below in Figure 6.

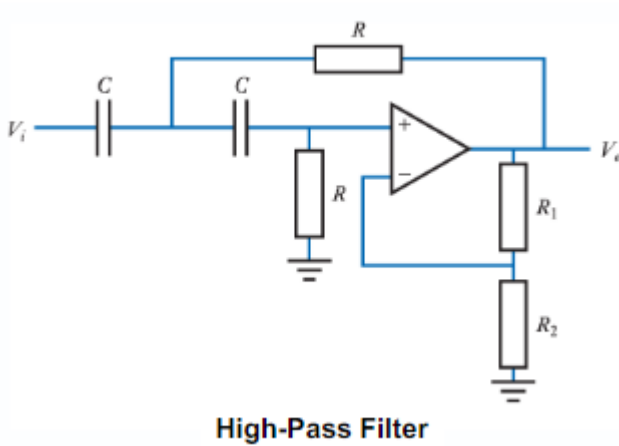


Figure 6. High-Pass Sallen-Key Filter Design [4].

Using this information, the team was able to design the circuit. Note that for the circuit above to be a Butterworth filter the Quality factor Q must be equal to $\frac{1}{\sqrt{2}} \cong 0.7$.

2 Circuit Design

To design the circuit the team had to consider what components were available to construct the design and what assumptions to make in order to reduce the complexity of the design.

2.1 Choosing Resistors and Capacitors based on Cut-off Frequency

The team decided to let resistors labeled R and capacitors labeled C in Figure 6 equal each other. Next the team decided to base the resistors (R) values off the available metal film capacitors. The relationship between these elements can be seen below in Figure 7.

$$f_H = \frac{1}{2\pi\sqrt{R_2 R_3 C_2 C_3}}$$

if

$$R_2 = R_3 = R, C_2 = C_3 = C$$

$$f_C = \frac{1}{2\pi RC}$$

This can be re-arranged to solve for R given a C:

$$R = \frac{1}{2\pi f_C C}$$

Figure 7. Relationship of Resistors (R) to Capacitors in a Butterworth Filter (adapted from source into Jupyter) [5].

The team put this relationship into a Python script (the full code was added to Appendix A) to analyze the possible resistance needed for each capacitor as shown below in Figure 8 and 9.

```

1 # assign constants
2
3 # quality factor
4 Q = np.sqrt(2)
5
6 # cut off frequency
7 f = 5500 #Hz
8
9 # metal film capacitors possibilities in F
10 existing_capacitors = [1*10**-8, 1.5*10**-8, 2.2*10**-8, 3.3*10**-8, 4.7*10**-8, 6.8*10**-8, 10*10**-8, 15*10**-8, 22*10**-8, 3
11
12

```

Figure 8. Python code of constants and available capacitors for the circuit design.

```

1 # Using the equation above for the freq and rearranging for R
2 def solve_R(c):
3     return 1/(2*np.pi * f*c)
4
5 #theoretical resistances:
6 resistors_T = []
7 for ele in existing_capacitors:
8     resistors_T.append(solve_R(ele))
9 print(str(resistors_T) + " in Ohms")
10
11 #Choose
12 print("\nTheoretical R: " + str(resistors_T[0]) + " Ohms with a Capacitance of " + str(existing_capacitors[0]) + " F") # note R
13
14 print("Actual R's: 2.7k + 180 = 2880 Ohms")

```

[2893.726238034461, 1929.1508253563072, 1315.3301081974823, 876.8867387983216, 615.6864336243533, 425.5479761815384, 289.3726238034461, 192.91508253563075, 131.53301081974823, 87.68867387983215, 61.56864336243534] in Ohms

Theoretical R: 2893.726238034461 Ohms with a Capacitance of 1e-08 F
Actual R's: 2.7k + 180 = 2880 Ohms

Figure 9. Python code of the suitable resistor values given each capacitor value.

From this analysis the group decided to go with the 0.01uF capacitors and theoretical resistors with 2894 Ohms. The actual resistors R in the physical circuit were made up of a 2.7k and 180 Ohm in series for a total of 2880 Ohms.

2.2 Gain of the Filter Based on Quality Factor

Now that the first part of the circuit is complete the team must ensure that the Op-Amp follows the Quality factor and Gain of a Second Order Unity Gain Buffer Butterworth Filter. To do this we must find a relationship for the output resistors R1 and R2 (voltage divider resistors) that satisfies the Quality factor and subsequent Gain of a Second Order Butter worth Filter. This relationship can be derived mathematically using the transfer function for a 2nd Order Butterworth Filter as shown below in Figures 10 and 11.

First write down the relationship between Gain and voltage in and out (A is Gain).

$$V_0 = AV_A$$

Note the Gain for a Unity Gain Buffer Amplifier

$$A = 1 + R_1/R_2$$

Writing the transfer function including a dampening factor for a Second Order Butterworth Filter:

$$\frac{V_0(s)}{V_i(s)} = \frac{A}{s^2 + 2Q\omega_n s + \omega_n^2}$$

A = overall gain,

Q = damping of system or Quality Factor,

ω_n = natural frequency of oscillations

$$\omega_n^2 = \frac{1}{R_2 R_3 C_2 C_3}$$

Solving the Transfer Function: When $R = R_1 = R_2$ and $C = C_1 = C_2$

$$\frac{V_0(s)}{V_i(s)} = \frac{A}{s^2 + \frac{3-A}{RC}s + \frac{1}{R^2 C^2}}$$

Figure 10. Transfer Function of a 2nd Order Butterworth Filter (adapted from source into Jupyter) [5].

Note that the 's' represents the Laplace operator which we can make the substitution $s=j\omega$.

From the 2 Equations for the Transfer function one can solve for the Gain (A) but substituting the Quality Factor Q.

Solving for Gain:

The Quality Factor required is:

$$Q = \frac{1}{\sqrt{2}}$$

then

$$0.707 = \frac{3-A}{2}$$

$$A = 1.586$$

Therefore the Gain of the circuit is about 1.586 v/v

Now we can solve for the Salen-Key Voltage Divider Resistors Ratio using the Gain:

$$A = 1 + R_1/R_2$$

Therefore:

$$A - 1 = R_1/R_2$$

$$R_1/R_2 = 0.586$$

Figure 11. Solving for Gain and Resistance Ratio of the Filter (adapted from source into Jupyter) [5, 6].

Therefore, it can be seen that the filter should have a Gain of 1.586 and a Resistance ratio (for R_1/R_2) of 0.586. Note that this ratio was also confirmed using the Textbook Horowitz and Hill: The Art of Electronics as shown below in Table 2, where K represents the Gain and where Poles represents the order of the filter.

Table 2. Comparison of Gain Values for various filters and Poles [6].

Poles	Butterworth K	Bessel		Chebyshev (0.5dB)		Chebyshev (2.0dB)	
		f_n	K	f_n	K	f_n	K
2	1.586	1.274	1.268	1.231	1.842	0.907	2.114
4	1.152	1.432	1.084	0.597	1.582	0.471	1.924
	2.235	1.606	1.759	1.031	2.660	0.964	2.782
6	1.068	1.607	1.040	0.396	1.537	0.316	1.891
	1.586	1.692	1.364	0.768	2.448	0.730	2.648
	2.483	1.908	2.023	1.011	2.846	0.983	2.904
8	1.038	1.781	1.024	0.297	1.522	0.238	1.879
	1.337	1.835	1.213	0.599	2.379	0.572	2.605
	1.889	1.956	1.593	0.861	2.711	0.842	2.821
	2.610	2.192	2.184	1.006	2.913	0.990	2.946

Therefore, the group chose to use a 1k Ohm resistor for R2 and a 584 Ohm resistor for R1. Note that in the physical circuit R1 is comprised of a 560 Ohm resistor in series with a 22 Ohm resistor for a total of 582 Ohms.

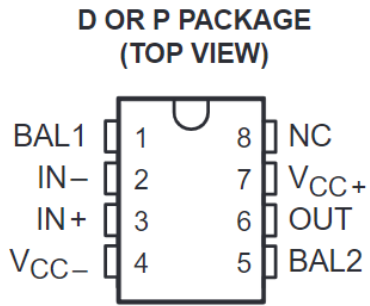
2.3 Final Design

To output the desired gain of 0.7 the group choose to include a voltage divider circuit consisting of an additional R1 and R2. A summary of the final design components can be seen below in Table 3.

Table 3. Summary of Circuit components.

Component	Theoretical / CAD Value	Physical / Real Value	Number Used
C	0.01uF	0.01uF	2
R	2894 Ohm	2880 Ohms	2
R1	584 Ohm	582 Ohm	2
R2	1000 Ohm	1000 Ohms	2

The team decided to implement this circuit using the LF411 OP-Amp as shown below in Figure 12.



NC – No internal connection

Figure 12. LF 411 Operational Amplifier Pin Outs [8].

The implementation of the LF411 Op-Amp can be seen in the OrCAD circuit diagram as seen below in Figure 13, including the pin outs and theoretical values of components.

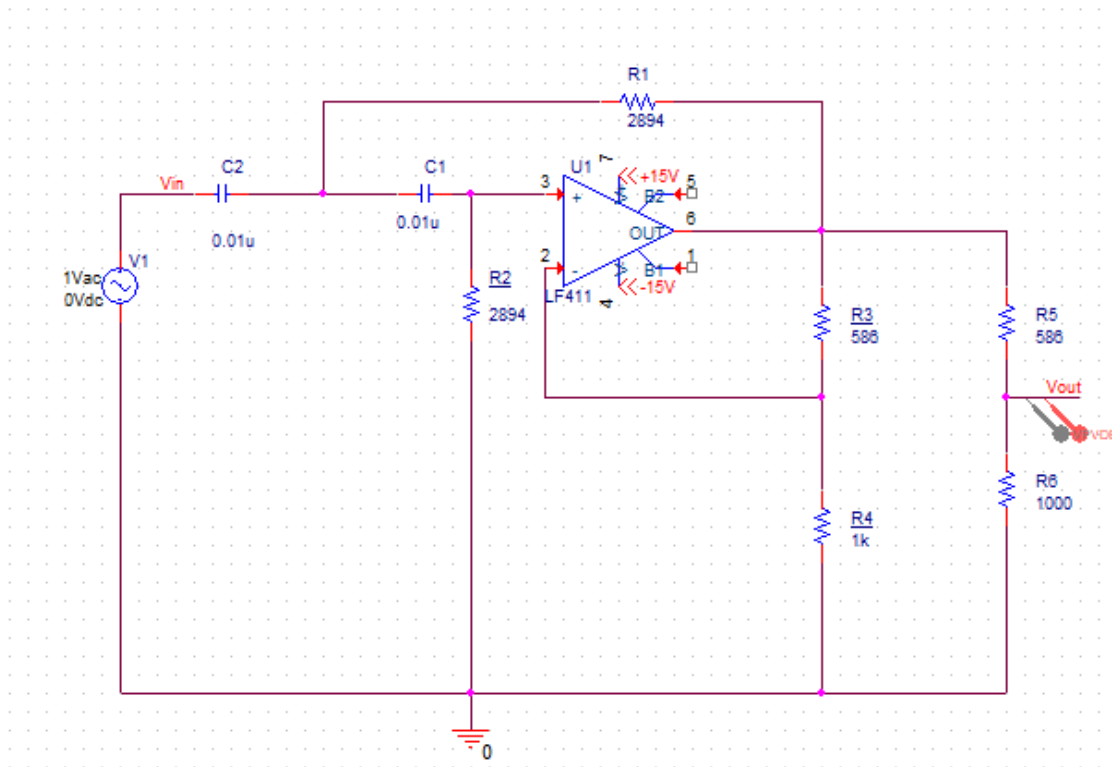


Figure 13. OrCad Final Circuit Design Schematic.

As well the physical set-up of our circuit can be seen below in Figure 14.

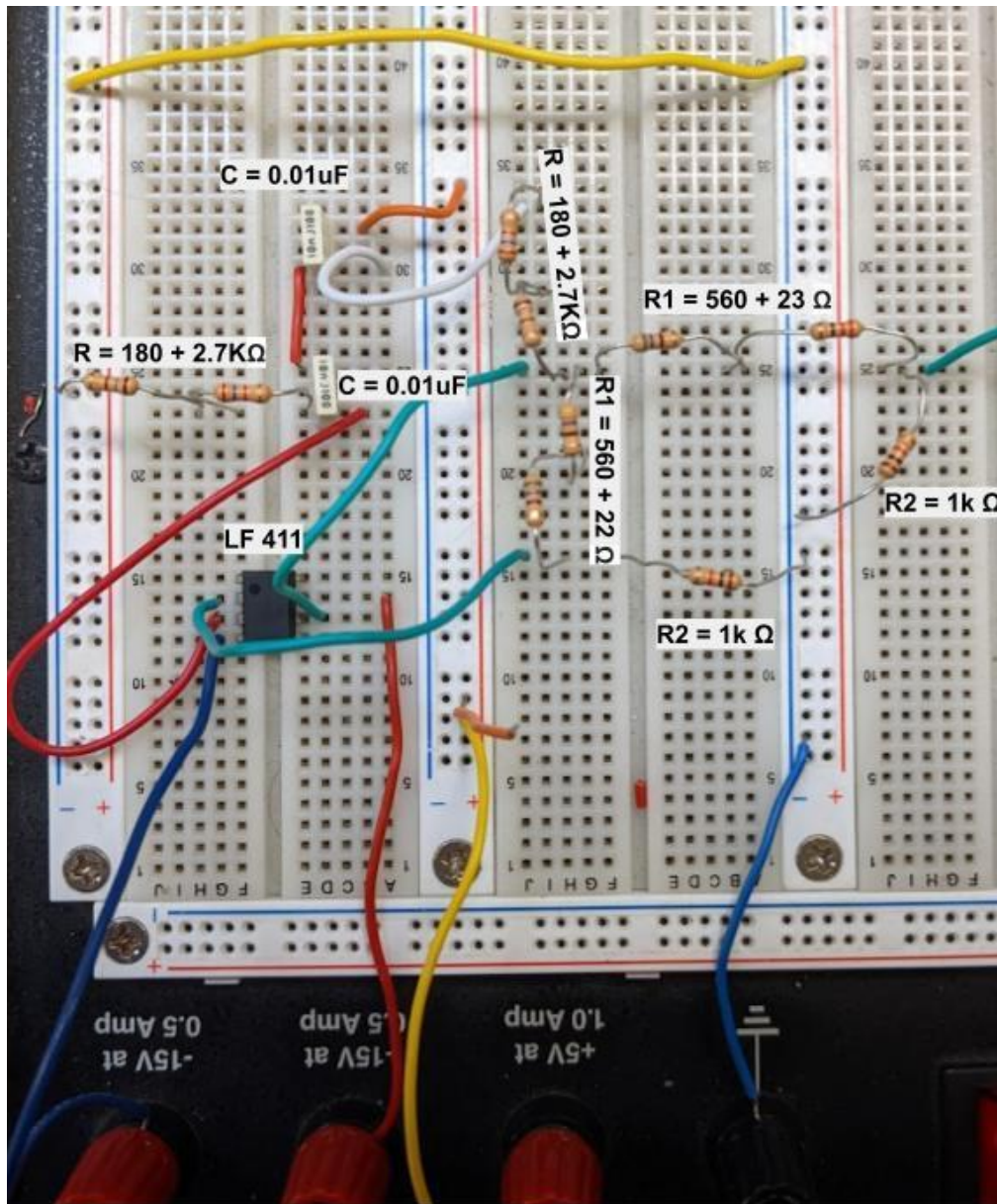


Figure 14. Physical Implementation of the Filter using the LF 411 Op-Amp.

Note that one of the R1 uses a 23 Ohm resistor (total 583 Ohms) while the other uses a 22 Ohm resistor total (582 Ohms) this is because of the group could only find one 23 Ohm resistor.

2.4 Results of The Final Design

From both the simulated and real circuit the design can be tested by observing the output results. From the CAD implementation the Bode Plot for Gain versus Frequency was produced as seen below in Figure 15.

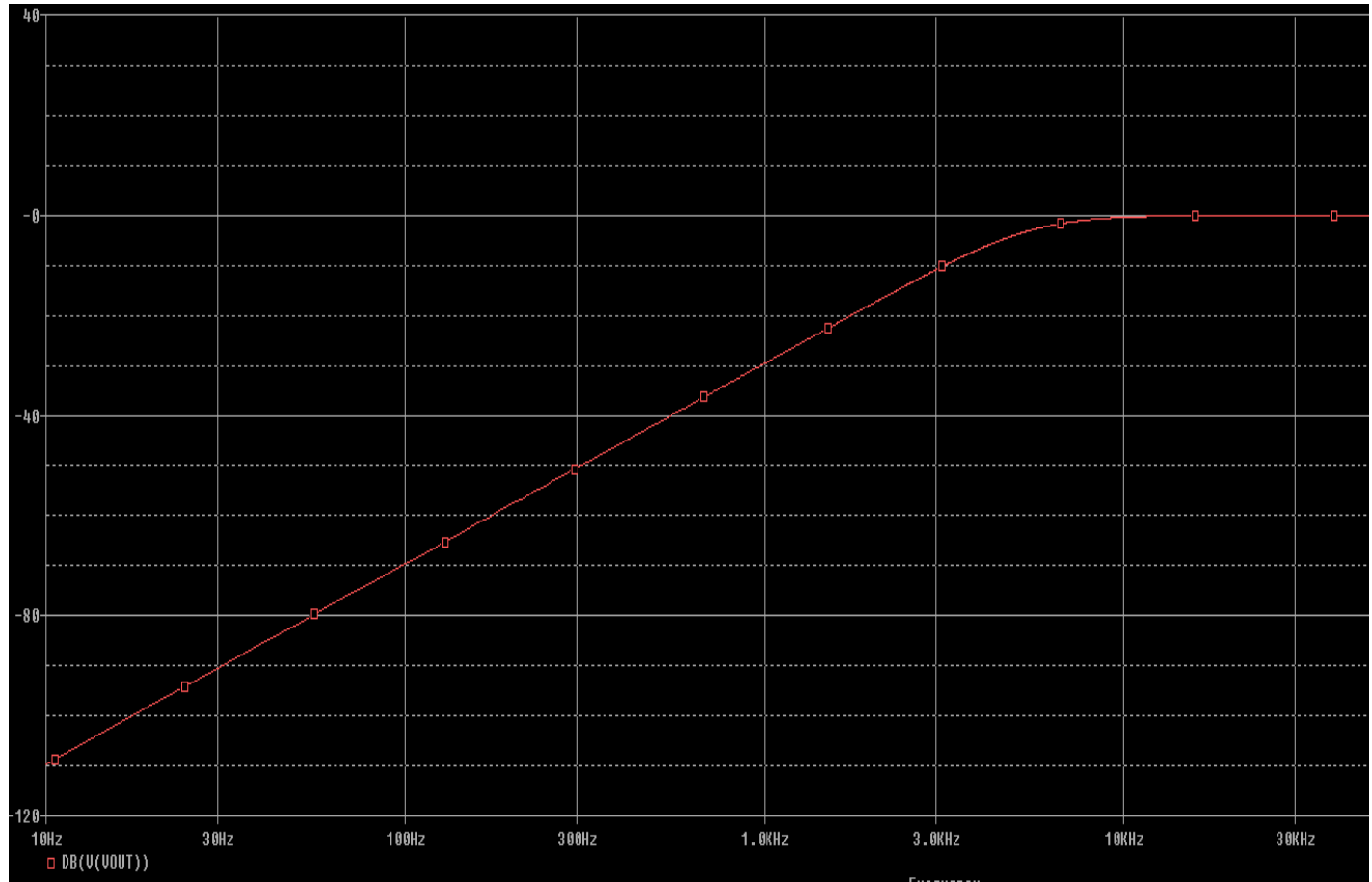


Figure 15. Gain in dB vs Frequency in Hz Simulated Bode Diagram.

This plot matches the expected output since the cut-off frequency is around 5500 Hz (when Gain in dB decreases 3dB from the pass band). As well the output along the pass band is around 0dB. As well the slope of the plot matches the gain in gain of 40 dB/decade. Further from 30Hz to 300Hz the plot goes from about -90 dB to -50 dB which matches 40 dB / decade. Also, the gain versus frequency plot shows a flat frequency response in the pass band which matches the output of a high pass Butterworth filter.

As well the Bode diagram for the Phase of the circuit versus frequency was produced as shown in Figure 16.

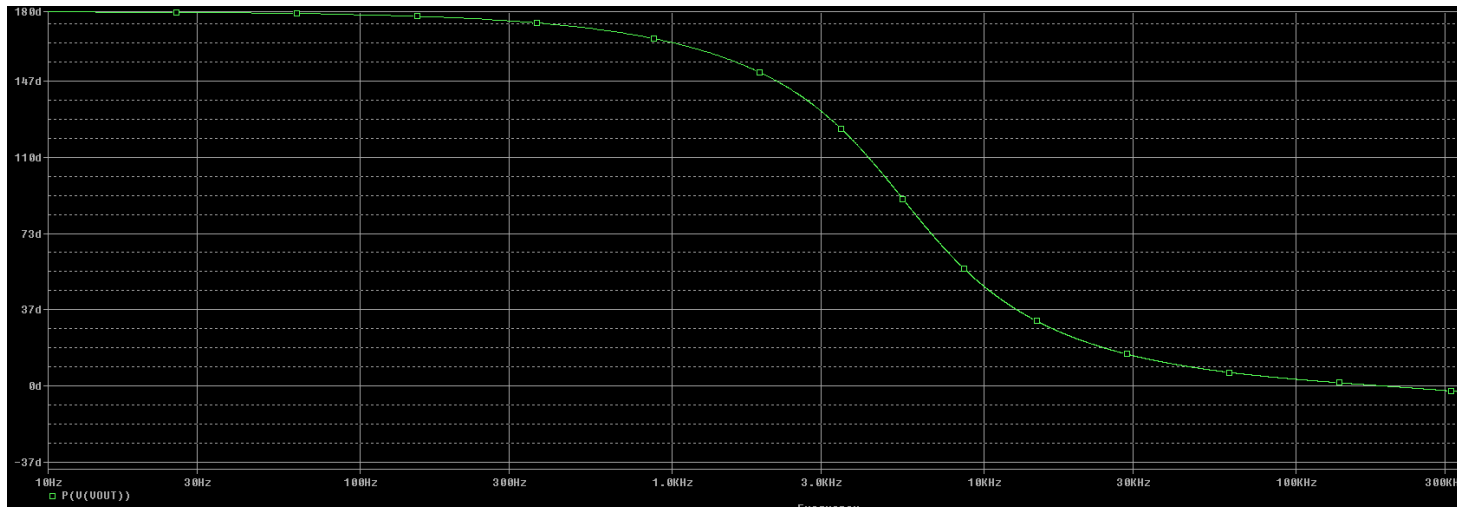


Figure 16. Phase Bode Diagram produce from OrCAD Circuit Analysis.

From the Phase plot we can see the Phase angle go from 180 to 0 degrees, this matches the output for a second order high pass network since it is double the amplitude of the 1st order plot. As well, the cut-off frequency can be found at exactly 90 degrees or at the middle of the phase plot curve.

The group also tested the physical circuit by passing a 1 Vpp (sinusoidal) analog voltage into the circuit and outputting the results onto the oscilloscope. The resulting waveform can be seen in Figure 17.

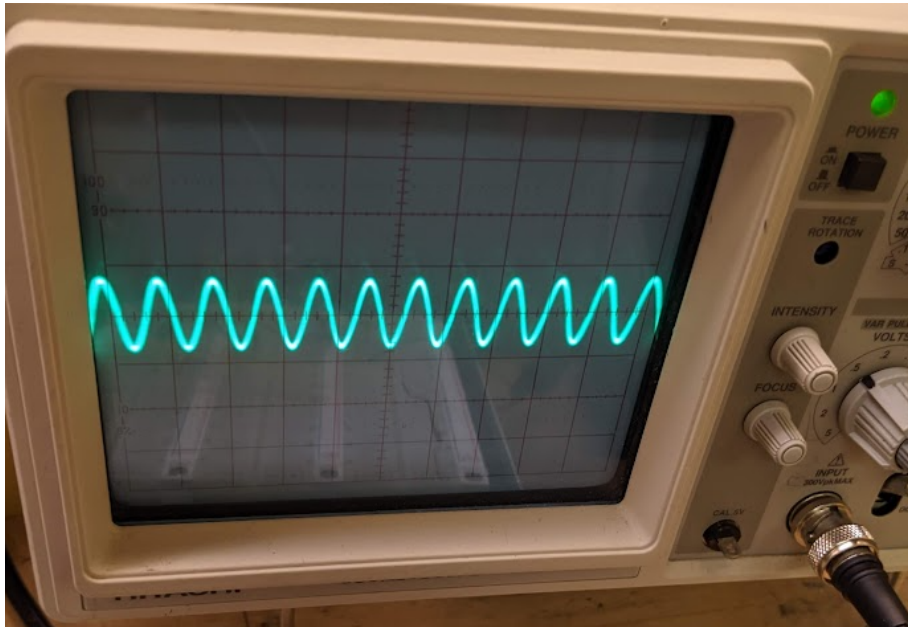


Figure 17. Resulting Voltage of the physical circuit.

As seen in Figure 17 the resulting amplitude matches the expected ~ 0.7 Vpp that the group expected to produce from the filter. These results show that the team was successful at building a 2nd Order Sallen-Key Butterworth filter with a cut-off frequency of 5500 Hz and a 40 dB change in gain.

References

- [1] B. Shastri, "Week 4 Frequency Response OP-Amps," Queens University, 2021. [Online]. Available: <https://onq.queensu.ca/d21/le/content/704814/viewContent/4200606/View>. [Accessed 25 October 2022].
- [2] Electrical 4 U, "High Pass Filter: Circuit, Transfer Function & Bode Plot," Electrical 4 U, 16 April 2021. [Online]. Available: <https://www.electrical4u.com/high-pass-filter/>. [Accessed 27 October 2022].
- [3] Circuits Today, "Active Filter Types," Circuits Today, 14 April 2011. [Online]. Available: <https://www.circuitstoday.com/active-filter-types>. [Accessed 21 October 2022].
- [4] B. Shastri, "Week 5 Filters," Queen's University, 2022. [Online]. Available: <https://onq.queensu.ca/d21/le/content/704814/viewContent/4200609/View>. [Accessed 26 October 2022].
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- [6] Electronics Tutorials, "Sallen and Key Filter," Electronics Tutorials, [Online]. Available: <https://www.electronics-tutorials.ws/filter/sallen-key-filter.html>. [Accessed 21 October 2022].
- [7] P. Horowitz, The Art of Electronics, Boston: Harvard University, 2015.
- [8] Texas Instruments, "LF411 JFET-INPUT OPERATIONAL AMPLIFIER," Texas Instruments, Dallas Texas, 1997.

Appendix A. Jupyter Code

In [1]:

```
1 import numpy as np
```

In [2]:

```
1 # assign constants
2
3 # quality factor
4 Q = np.sqrt(2)
5
6 # cut off frequency
7 f = 5500 #Hz
8
9 # metal film capacitors possibilities in F
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11
12
```

$$f_R = \frac{1}{2\pi\sqrt{R_2 R_3 C_2 C_3}}$$

if

$$R_2 = R_3 = R, C_2 = C_3 = C$$

$$f_C = \frac{1}{2\pi RC}$$

This can be re-arranged to solve for R given a C:

$$R = \frac{1}{2\pi f_C C}$$

In [3]:

```
1 # Using the equation above for the freq and rearranging for R
2 def solve_R(c):
3     return 1/(2*np.pi * f*c)
4
5 #theoretical resistances:
6 resistors_T = []
7 for ele in existing_capacitors:
8     resistors_T.append(solve_R(ele))
9 print(str(resistors_T) + " in Ohms")
10
11 #Choose
12 print("\nTheoretical R: " + str(resistors_T[0])+" Ohms with a Capacitance of " +str(existing_capacitors[0]) + " F") # note R
13
14 print("Actual R's: 2.7k + 100 = 2800 Ohms")
```

[2893.726238034461, 1929.1588253563072, 1315.3301081974823, 876.8867387983216, 615.6864336243533, 425.5479761815104, 289.3726238034461, 192.91588253563075, 131.53301081974823, 87.68867387983215, 61.56864336243534] in Ohms

Theoretical R: 2893.726238034461 Ohms with a Capacitance of 1e-08 F

Actual R's: 2.7k + 100 = 2800 Ohms

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A = overall gain,

Q = damping of system or Quality Factor,

ω_n = natural frequency of oscillations

$$\omega_n^2 = \frac{1}{R_2 R_3 C_2 C_3}$$

Solving the Transfer Function: When $R = R_1 = R_2$ and $C = C_1 = C_2$

$$\frac{V_O(s)}{V_I(s)} = \frac{A}{s^2 + \frac{3-A}{RC}s + \frac{1}{R^2 C^2}}$$

Solving for Gain:

The Quality Factor required is:

$$Q = \frac{1}{\sqrt{2}}$$

then

$$0.707 = \frac{3-A}{2}$$

$$A = 1.586$$

Therefore the Gain of the circuit is about 1.586 v/v

Now we can solve for the Sallen-Key Voltage Divider Resistor Ratio using the Gain:

$$A = 1 + R_2/R_1$$

Therefore:

$$A - 1 = R_2/R_1$$

$$R_1/R_2 = 0.586$$

we decided to use a 1k for R_2 Therefore $R_1 = 588$ Ohms theoretically:

The Group used these values for Resistors

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